Outline

1. Relaxion
   (i) QCD Axion Review
   (ii) "Relaxing" the Higgs Mass
2. Extra Spacetime Dimensions
   (i) Flat extra dimensions
   (ii) Randall-Sundrum models

(* Do heirarchy problem review at 4 sharp *)

Dirac Naturalness: In an EFT, dimensionless parameters are $O(1)$.

Problem: $\underset{O(100\,GeV)}{m_H^2} = c_0 \Lambda^2 \leftarrow$ cutoff of SM

$\Lambda \sim M_{pl} \sim O(10^{18}\,GeV)$

$\rightarrow c_0 \ll 1$ :(

t'Hooft (technical) naturalness: Small dimensionless parameters are allowed if the symmetry of your theory is enhanced when $c_0 \rightarrow 0$

No such symmetry protects the Higgs mass

(* Do outline afterwards *)

Why do QCD Axion review?

1. The relaxion is an axion!

Strong CP Problem

CP: particles $\rightarrow$ antiparticles

2 Sources of CP Violation in Strong Sector:

1) Term in $\mathcal{L}_{QCD} \supset \frac{\alpha_s}{8\pi} \theta_{QCD} G_{\mu\nu} \hat{G}^{\mu\nu}$

bare coupling

$$\hat{G}^{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} G_{\rho\sigma}$$

dual gluon field strength

(i) Total derivative, but with physical significance
(ii) $\theta$ is an angle
(iii) CP odd

→ total derivative (if $A^{\mu}_a$ vanishes @ $\infty$, this term would not matter
→ QCD has a nontrivial topological vacuum structure
→ have multiple unique vacua which different topology labeled by winding number $n$ → describe # of times gauge field configuration wraps around SU(3) gauge group (topological invariant)
→ $\theta$ parametrizes overlap between many degenerate vacua

2) complex phase in weak sector CKM matrix

field redefinitions → $\mathcal{L} \supset \frac{\alpha_s}{8\pi} \theta G_{\mu\nu} \hat{G}^{\mu\nu}$

$$\theta = \theta_{QCD} + \theta_{weak}$$

Experimental constraints on neutron EDM put

Experimentally: $\theta \lesssim 10^{-11}$

Fine tuning problem!

**Axion:**

(Kim, Shifman, Vainshtein, & Zakharov (KSVZ))

$\sigma$ : - complex scalar
- singlet under SM

$Q$ : - quark
- charged under $SU(3)_c$

$$\mathcal{L}_{KSVZ} \supset y\sigma \bar{Q}_L Q_R + h.c.$$

$\mathcal{L}_{KSVZ}$ has a $U(1)_A$ (Axial) PQ symmetry

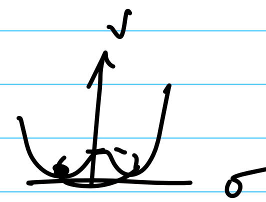
$$Q_R \to e^{i\alpha} Q_R$$
$$Q_L \to e^{-i\alpha} Q_L \qquad (\bar{Q}_L \to e^{i\alpha} Q_L)$$
$$\sigma \to e^{-2i\alpha} \sigma$$

PQ = Peccei-Quinn

Put $\sigma$ in a (Mexican-hat) potential

$$\mathcal{L}_{KSVZ} \supset -\lambda_\sigma \left(|\sigma|^2 - v_{PQ}^2/2\right)^2$$

$$v_{PQ} \gg \Lambda_{EW}$$




Vacuum will settle into the minimum of the potential. The $U(1)_A$ symmetry is spontaneously broken. To study excitations about this vacuum, makes sense to reexpand our field

$$\sigma = \frac{1}{\sqrt{2}}\left(v_{PQ} + \rho(x)\right) e^{i a(x)/v_{PQ}}$$

axion!

$\rho$, $Q$ will pick up $m \sim v_{PQ} \gg \Lambda_{EW}$

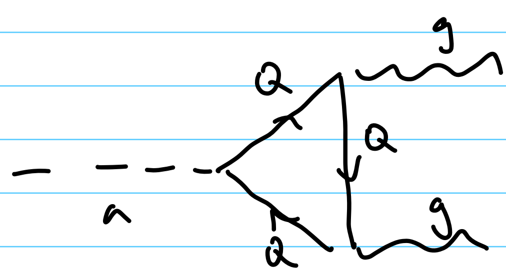
$a(x)$ :
- massless (Goldstone's theorem)
- compact field $a(x) \in [0, 2\pi v_{PQ}]$
- $\mathcal{L}$ symmetric under $a(x) \to a(x) + c$, $c \in \mathbb{R}$

Integrate massive fields out of theory

$$\mathcal{L}_{EFT} = \partial_\mu a \partial^\mu a$$

Anomaly: Classical Symmetries $\neq$ Quantum Symmetries

Anomaly ($\Delta$) Diagrams




In the limit $M_Q \gg \Lambda_{EW}$

$$\mathcal{L}_{EFT} \supset \frac{1}{2}\partial^\mu a \, \partial_\mu a + \frac{\alpha_s}{8\pi}\frac{a}{f_a} G^a_{\mu\nu}\tilde{G}^{a\mu\nu}$$

$f_a = v_{PQ}$

Old Symmetry: $a(x) \to a(x) + c$

New Broken "Symmetry": $a(x) \to a(x) + 2\pi k f_a$, $k \in \mathbb{Z}$

(Not really a symmetry)

$$a' = a + f_a(\theta_{QCD} + \theta_{weak})$$

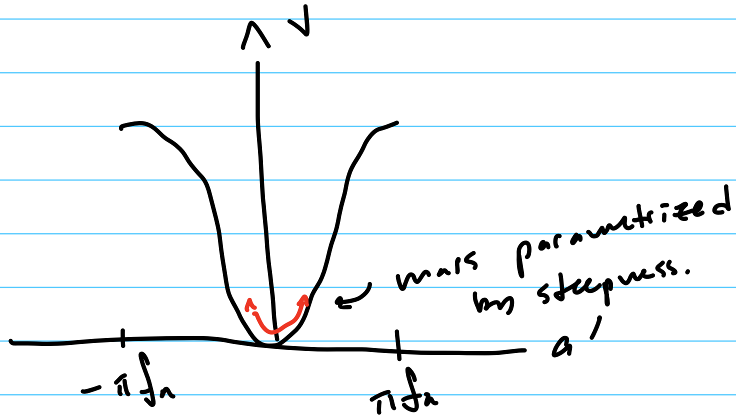
$$\mathcal{S} \supset \underbrace{\frac{\alpha_s}{8\pi}\frac{a'}{f_a} G^a_{\mu\nu}\tilde{G}^{a\mu\nu}}_{\text{All Strong CP violation}}$$

QCD vacuum induces nontrivial potential for the axion after EWSB / QCD confinement

$$\Lambda^4_{QCD} \sim f_\pi^2 m_\pi^2 \sim y v f_\pi$$

$v$: Higgs vev

$$V \sim -\Lambda^4_{QCD} \cos(a'/f_a)$$



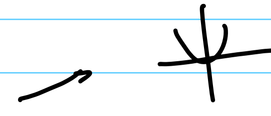

minimized for $\langle a' \rangle = 0$!

Classically, $a' = 0$. Quantum mechanically, excitations above vev are physical axion particles you can measure

Relaxion

Goal: Devise a mechanism by which $m_H^2$ can be tuned to an unnaturally small value, dynamically.

$$\mathcal{S} \supset \partial_\mu\phi\,\partial^\mu\phi + \frac{\alpha_s}{8\pi}\frac{\phi}{f}\tilde{G}^{\mu\nu}G_{\mu\nu} + V(g\phi) - (\underbrace{\mu^2 - g\phi}_{m_H^2 > 0})|H|^2$$

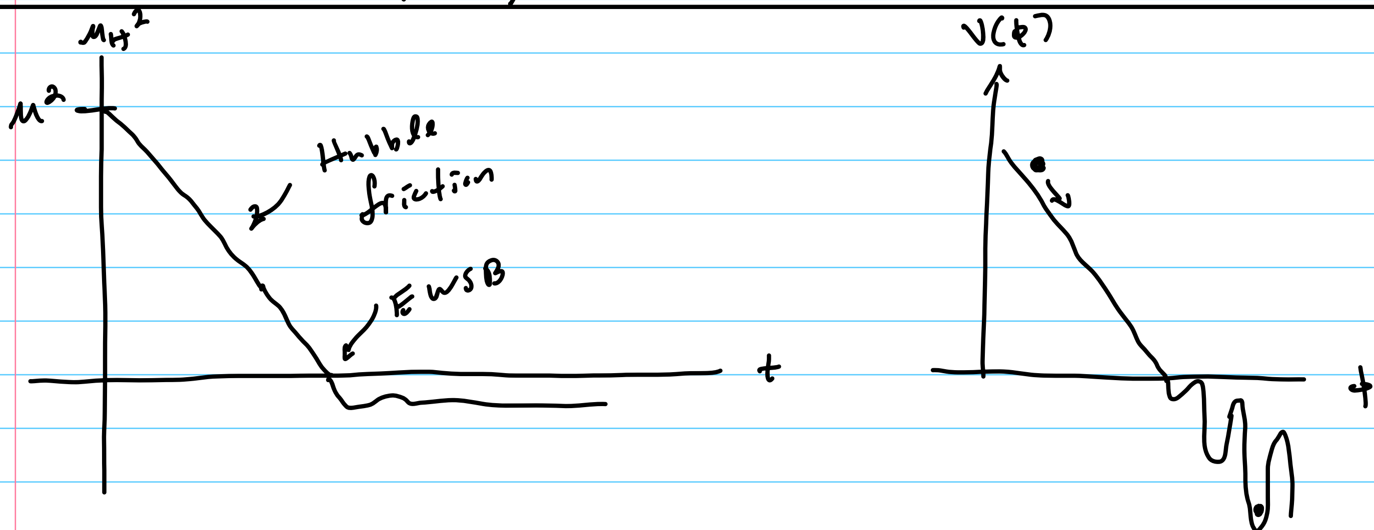


- axion-like particle $\phi$, $\phi \sim O(1)$ to start
- potential

$$V(g\phi) = g\mu^2\phi + g^2\phi^2 + \cdots$$

that breaks discrete shift "symmetry" according to $g/\mu$
- Couple to Higgs, $\mu \sim \Lambda$

$\phi$ is not compact, takes values $\gg f$




→ $V(\phi)$ pushes $\phi$ to increase, $m_H^2$ to decrease
→ Need Hubble friction to dissipate $\phi$'s KE (must occur during inflation)
→ EWSB! $V(\phi)$ & $\cos(\phi/f)$ compete
→ $\phi$ stops $\sim$ $g \sim \dfrac{y v f_\pi^3}{\mu^2 f}$

Allows $m_H \ll \Lambda$

$$\mu = 10^7\,\text{GeV},\ f = 10^9\,\text{GeV} \to \underbrace{g/\mu \sim 10^{-30}}_{\text{small}}$$

Problems:

1) This does not fix strong CP $\to$ $\theta_{QCD} \sim O(1)$
Solution: → couple axion to a different $SU(3)_N$ not in SM (new axion)

2) Worsens fine tuning for cosmological constant!

→ How much does physics depend on which minima $\phi$ settles in?

- $\Delta m_H^2 \sim g f \sim \Lambda^4_{QCD} / \mu^2$ small!
- $\Delta V \sim g f \mu^2 \sim \Lambda^4_{QCD}$ large compared to c.c.

3) Not ('t Hooft) technically natural

→ $g/\mu$ does not break a real symmetry

→ $g/\mu$ is not "protected" up to scale $\Lambda$

→ to restore discrete shift "symmetry"

$f > \Delta\phi \sim \mu^2/g$.

→ cosmology tells us $\mu \sim 1/fp$ → $\mu \sim \Lambda_{EW}$ :(

Solution: Don't rely on inflation

## Extra small Spacetime dimensions

- SM lives on 4D slices of ND space (branes)
- Graviton describes dynamics of $g_{\mu\nu}$
  → lives on full Ndim space

### Flat

Ndim Graviton is massless

$$P = (p_0, \vec{p}_3, \vec{p}_E)$$

$$P^2 = 0 \to -p_0^2 + \vec{p}_3^{\,2} = \underbrace{-\vec{p}_E^{\,2}}_{} \equiv -m_4^2$$

graviton, generally looks massive in 4D

→ 4D Graviton (appears massless in 4D) is mode $\vec{p}_E = 0$

Classically, $S_{EH} = \int d^4x\, \underbrace{M_{pl}^2}_{} \sqrt{-g_4}\, R_4(g_4)$

$$S_N = \int d^N x\, \Lambda^{N-2} \sqrt{-\tilde{g}}\; R(\tilde{g})$$

If we can relate $S_{EH}$ to $S_N$ we can relate $M_{pl}$ to $\Lambda$

"Conformally flat" metric

$$\boxed{ds^2 = e^{A(x^m)} \left( \underbrace{g_4^{\mu\nu}(x_\mu)}_{} dx_\mu dx_\nu + \sum_m dx_m^2 \right)}$$

uniform warp factor

4D gravitational fluctuations

$$\sqrt{-\hat{g}} \longrightarrow e^{A(x^m)\cdot N/2}\sqrt{-g_4}$$

$$R(\hat{g}) \longrightarrow e^{-A(x^m)} R_4(g_4) + (\underbrace{\text{derivatives of } A}_{\text{do not contribute to 4D gravity}})$$

$$S_N = \int d^4x \, \Big(\underbrace{\int d^{N-4}x \; \Lambda^{N-2} \, e^{A(x^m)\cdot \frac{N-2}{2}}}_{M_{pl}^2}\Big) \sqrt{-g_4}\, R_4(g_4)$$

Assumptions:

(i) Flat $\rightarrow$ $A(x^m)=0$

(ii) All extra dimensions are size $r_0$

$$M_{pl}^2 = \Lambda^2 \cdot (\Lambda r_0)^{N-4}$$

$\Lambda \sim 1\,\text{TeV}$

$N=5$ : $r_0 \approx 5\text{AU}$ ✗ already been probed

$N=6$ : $r_0 \sim \text{mm}$ ✓ just being probed now

This solution works by lowering the cutoff such that the Higgs mass is of natural size.

New problem: Why is $\Lambda r_0$ so large

e.g. in $N=6$ : $1/r_0 \sim 0.1\,\text{meV}$ $\longrightarrow$ new heirarchy problem

# Randall - Sundrum model

**Horizon problem:** Universe appears homogenous and isotropic but opposite points on the horizon appear as if they have never been in causal contact.

**Solution:** Inflation!

$$ds^2 = -dt^2 + e^{\sqrt{\pm 2\alpha/3}\, t} d\vec{x}^{\,2}$$

$\longrightarrow$ c.c. $= \alpha$ (de-Sitter geometry

As time progresses, proper distance scales become exponentially warped

Let's try this ourselves

$$ds^2 = e^{2ky}\left(-dt^2 + d\vec{x}^{\,2}\right) + dy^2 \quad \swarrow 5D$$

$$\text{c.c.} = -6k^2$$

As we move along the extra dimension, proper distance & thus mass scales will become exponentially warped!

Suppose Higgs Boson brane lives at $y = y_0$

$$S = \int d^4x\, dy\, \delta(y-y_0)\sqrt{-g}\left(g^{\mu\nu}\partial_\mu H^\dagger \partial_\nu H - \lambda\left(|H|^2 - f^2\right)^2\right)$$

$$f \sim \Lambda$$

$$= \int d^4x\, e^{-4ky_0}\left(e^{2ky_0}\eta^{\mu\nu}\partial_\mu H^\dagger \partial_\nu H - \lambda\left(|H|^2 - f^2\right)^2\right)$$

Field redefinition

$$H' = e^{-ky_0} H$$

$$= \int d^4x \left( \eta^{\mu\nu} \partial_\mu H^\dagger \partial_\nu H - \lambda\left(|H|^2 - (f e^{-ky_0})^2\right) \right)$$

$$m_H \sim v \sim \lambda e^{-ky_0}$$

$$y_0 \sim \frac{1}{k} \log \frac{\Lambda}{m_H}$$

e.g. $\log \frac{m_{pl}}{m_H} \sim 37$

True Solution! No new heirarchy problem since $r_0$ need not be heirarchically larger than other length scales

Problem: We do not live in an AdS universe.

To solve Einstein's equations, must account for the cosmological constants taken on by 4D boundaries of extra dimension. Tuning these constants, we can live in a 4D universe with an appropriately small cosmological constant